

What Are UX and UX Design?

Fine art and pizza delivery: what we do falls neatly in between.

—David Letterman

Highlights

- Definition of UX
- Pervasiveness of computing
 - Evolution of computing
 - Expanding concept of interaction
- The broad scope of UX
- Designing for a good user experience
- The components of UX
 - Usability
 - Usefulness
 - Emotional impact
 - Meaningfulness
- Evolving importance of UX
- A business case for UX
- Brief history and roots of HCI and UX

1.1 UX IS SHORT FOR USER EXPERIENCE

Clearly, this whole book is about UX, so we need to get right to the point about what UX is. As we said in the Preface, the two letters UX are a popular acronym that stands for user experience. Those two letters stand for the whole field, all the work that is done in this field, and the final user experience that comes out of that work. See Section 1.8 for some history related to human-computer interaction (HCI) and UX.

1.2 THE ACCELERATING TREND OF PERVASIVE COMPUTING

In the second edition, we talked about how the trend of computing ever pervades deeper into our lives. Now, in the third edition, we find this trend accelerating. So-called smartphones are continuing to get smarter and more capable of supporting more and more of our needs—they are now enabling the way we pay at checkout counters; the way we authenticate ourselves (fingerprints, face detection) on products; the way we navigate through bus, rail, and air transit systems; and more. Wearable devices, such as smartwatches, are becoming more useful with great advances in health monitoring and are even saving lives (irregular heart rhythm, sleep quality, accident and fall detection). Computing has pervaded into our cars, making them ever so closer to being autonomous.

1.2.1 Evolution of Computing and Interaction

For a long time, interaction was about screens full of data fields displayed on dumb terminals connected to mainframes. When personal computers first came along in the late 1970s, their screens mostly looked the same—countless panels containing data fields. Then, along came graphical user interfaces (GUIs); the internet and web; and personal, handheld, and mobile devices as well as devices embedded in appliances, homes, and cars. We went from “doing computing” to using devices for living our lives in interactive environments. As Mark Weiser (1991) said, “The world is not a desktop.”

Disappearing technology. Interaction, however, is doing more than just reappearing in different devices such as we see in web access via a mobile phone. Weiser (1991) stated that “the most profound technologies are those that disappear.” Russell, Streitz, and Winograd (2005) also talked about the disappearing computer—not computers that are ceasing to exist, but those that are becoming unobtrusive and unremarkable by disappearing “into the walls and interstices of our living and working spaces.”

1.2.2 The Expanding Concept of Interaction

As a result of changes in the way computing is viewed, the notion of interaction has evolved into a broad concept, referring to a wide variety of communication and collaboration between a human and an interaction artifact, such as a device, product, or system in an ecology.

New interaction artifacts. An interaction artifact is a system, device, service, instrument, mechanism, object, or environment that can communicate with

a human in either or both directions. As we expand our perspective from the traditional desktop computer, an interaction artifact, then, can be the building or room you are in, the chair you may be sitting in, an automatic teller machine, an elevator, an appliance such as a refrigerator, a car or other vehicle, most kinds of signage, a home, the workflow of the Department of Motor Vehicles, or a voting machine.

New modes of interaction. It is not just the variations of devices (artifacts) that have changed but the nature of interaction itself, which has expanded from the use of a keyboard, mouse, and monitor to handheld devices such as touch-sensitive smartphones. Now, smartwatches have changed the nature of interaction to a combination of touch and scroll wheel—and smartglasses, virtual or augmented reality goggles, and smart living environments let users interact using yet other modalities, such as voice.

The nature of interaction now spans multiple human sensory dimensions, with newer patterns suitable for those dimensions: from the traditional WIMP (windows, icons, menus, pointers) mode of interaction to touch, from a predominantly graphical medium of interaction to voice and vision, and from users translating their goals to specific actions on a two-dimensional system to direct interpretation of brain activity by the system via probes on the human brain.

Because interaction is so broad and pervasive in our world, good UX design will necessarily have a substantial positive impact on our lives.

1.2.3 UX Across the User's Ecology

Not all interaction is for a single user to perform a simple task, such as adding an item to a calendar. Rather, an interaction can continue through different states in time and space and through different environments.

1.2.4 Activity-Based Interaction

Norman (2005) introduced activity-based design as a way to describe interactions that go beyond simple tasks. He describes activity-based interaction as a set of multiple, overlapping, and related task threads all serving a common work goal within the user's ecology.

Activity-based interaction on one device. Activity-based interaction can occur on a single device. As an example, suppose you are searching online for a compact digital camera. You might follow links to reviews, decide on one, put it in your e-cart, then follow links to other similar products. You can also follow other links to accessories (e.g., SD memory card, camera case, wrist strap, USB

cable for downloading) you might want, and so on. Even though this involves multiple different tasks, users think of it as doing one activity.

Activity-based interaction across multiple devices. Activity-based interaction can also span devices. As an example, consider how people use smartwatches, such as the Apple Watch. This example involves multiple devices and tasks, but users center on one activity: managing their watch. Apple Watch is designed to have its own surrounding ecology to support several related activities. For example, you might use an iPhone to update your Apple Watch. If you want to install a new app on the watch or customize your notifications, then you have to do all that on your iPhone.

Activity-based interaction across devices in the user's ecology. System-spanning interaction often involves multiple parties in multiple work¹ or play roles, multiple devices, and multiple locations, embracing the user's whole ecology. This theme of including the broader ecology runs through the entire book.

Example: Powerlines Are Down

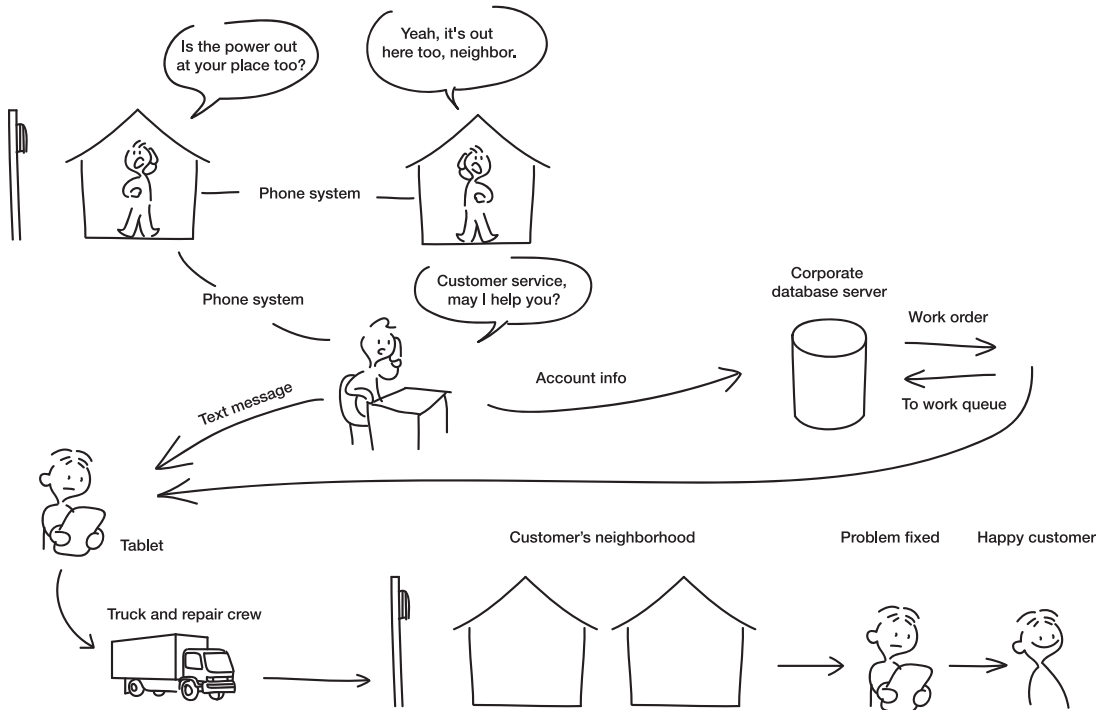
Here is an example adapted from Muller, Wildman, and White (1993a) of a system-spanning, activity-based transaction with a relatively simple goal of getting electrical power service restored to a user who finds the power is out in her house.

The activity begins when our user calls her neighbor (Fig. 1.1). The neighbor says his power is out, too, and he thinks a powerline is down in the neighborhood. Our hero then calls the power company customer service line and asks if it can be fixed. The customer service person posts a work order in a queue in a central database, with account number, customer name, phone number, and address.

Customer service also sends a text message to a technician in the field, who checks the queue on his portable tablet and picks up the work order, hops in his truck, and drives to that neighborhood. The technician and crew fix the powerlines, and the technician then reports the job completed on his portable tablet. The customer is happy.

From this one simple example, you can see an ecology involved in one user activity that includes the user as customer, the power company, its customer service, customer accounts, the power company's central database, a work order queue, a work order, power company field technicians, and the powerlines. The ecology of this activity also includes the telephone system, the neighbor on his phone, a text message, the technician's portable tablet, the technician's crew, and a fleet of power company trucks.

¹When we use the term *work*, we intend it to broadly signify anything people/users do while using products or systems. See Section 6.2.



*Fig. 1.1
System-
spanning
activity to
fix downed
powerlines.*

The user's workflow is a continuum across that ecology, and information accessibility across the whole ecology is the glue that ties the different sub-systems within the ecology together.

1.3 ENTER UX

In response to these trends, we find HCI and usability of earlier times also expanding—into UX.

1.3.1 A User Experience Is Felt by the User

A user experience is the totality of the effects felt by the user before, during, and after interacting with a product, system, or other parts of an ecology (Fig. 1.2). In the figure, we summarize the concepts of interaction and UX springing from that interaction.

Breaking out the key characteristics reflected in this definition, a user experience:

1. Is a result of interaction
2. Is about the totality of the effects
3. Is felt internally by a user
4. Includes usage context and ecology

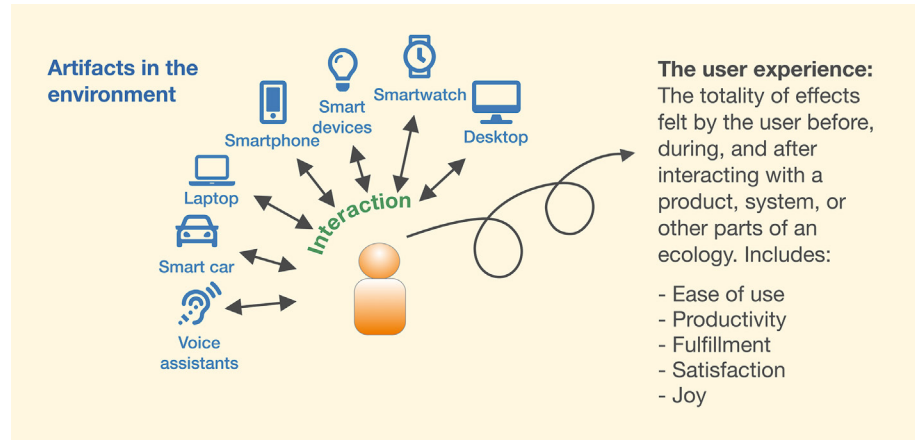


Fig. 1.2
Human-artifact
interaction and the
resulting user experience.

Results from interaction. It is obvious, but must be stated, that a user experience results from interaction. Here, we use the term *interaction* in the broadest sense, including seeing a marketing message long before direct physical interaction with the product to memories of those encounters long after it ceases to exist.

Totality of effects. Following up on the second characteristic of a user experience, as the Dagstuhl (2011) report states, the effects of interaction include the user's entire "stream of perceptions, interpretations of those perceptions, and resulting emotions during an encounter with a system." Since then, the concept of "an encounter" has broadened to include emotion arising before, during, and after physical contact with the product.

That totality of effects of interaction includes:

- Influence of usability, usefulness, and emotional impact during interaction
- Full unfolding of these effects over time

As an example of effects felt over time, consider a potential user researching a product or system, seeing advertising and reviews, and anticipating ownership. Once the product is bought, the effects include product packaging and an out-of-the-box experience; seeing, touching, and thinking about the product; and admiring the product, using it, and retaining and savoring (or not) the pleasure of usage.

Lastly, a user experience can include the individual's feeling about the company that produced the product or system and its reputation and branding, as well as the pride of ownership and how the product has acquired meaning in the user's lifestyle, extending into a broad cultural and personal experience.

A user experience is felt internally by the user. Clearly, it is the user who has the experience. Therefore, user experiences from interaction under the same conditions can vary across individual users.

Context and ecology are crucial to a user experience. An ecology is the complete usage context, including all parts of the world the user comes in contact with related to the interaction. The user can be part of multiple ecologies (e.g., work vs. home). Within an ecology, there could be multiple specific usage contexts (e.g., stressful work conditions or pleasurable play conditions), and each such context affects the user experience.

Distinction from HCI. Human-computer interaction is an earlier term for the whole field of interest. In some way, then, UX is a subset of HCI. In the general public, however, the terms *usability*, *HCI*, and *UX* are somewhat interchangeable in reference to the discipline or field of interest. In this book, we take the perspective of UX being the practice of designing for a high-quality user experience.

1.3.2 Can a User Experience Be Designed?

The perceptive reader may have already spotted a small inconsistency. We sometimes see the phrases “UX design” and “designing a user experience.” But you cannot design something that occurs internally in a user. So, phrases such as “UX design” really do not make sense, but we trust you will understand that this means designing *for* the user experience.

1.4 COMPONENTS OF A USER EXPERIENCE

A user experience can include how users experience four factors (Fig. 1.3):

1. Usability
2. Usefulness
3. Emotional impact
4. Meaningfulness

1.4.1 Usability

Long ago, the field of HCI, the umbrella academic discipline for UX, was primarily about usability, which includes (ISO 9241-11, 1997):

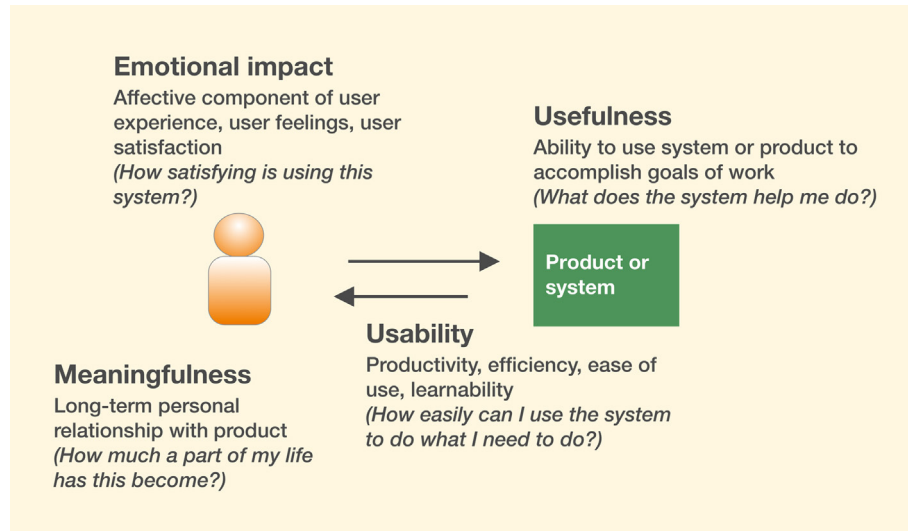


Fig. 1.3

The components of a user experience include usability, usefulness, emotional impact, and meaningfulness.

- Ease of use
- User performance and productivity
- User satisfaction
- Efficiency
- Error avoidance
- Learnability
- Retainability (ease of remembering)

The broadening of focus to UX has not supplanted usability, which remains an important part. As the field focuses more on the glamorous parts of the user experience, however, sometimes the foundational component—usability—has been forgotten. Without good usability, it is rare that the other components of a user experience will be considered.

1.4.2 Usefulness

The usefulness component is about utility. Usefulness speaks to the power and functionality of the backend software that gives you the ability to accomplish work (or play). It is the underlying reason for the existence of a product or system.

1.4.3 Emotional Impact

The third component is emotional impact, an affective part of a user experience. As the term implies, emotional impact embraces how users feel about an interaction (Norman, 2004), including user satisfaction.

Although there were earlier academic papers about emotion in the user experience, Norman (2002) was one of the first to bring the topic to light on a broad scale, relating it to his theme of everyday things.

Technically, the totality of a user experience is emotional because it is all experienced internally by the user; however, some user experience factors are more purely emotional, are felt up close and personal during interaction, and take the user beyond simple satisfaction to fun, enjoyment, and self-expression—sometimes with strong emotional consequences.

Emotional impact can be experienced in many ways:

- Joy of usage
- Pleasure
- Excitement
- Fun
- Curiosity
- Aesthetics
- Novelty
- Surprise
- Delight
- Exploration
- Coolness
- Appeal
- A sense of identity
- Happiness
- Enthusiasm
- Enticement
- Engagement
- Pride of ownership
- Affinity, attractiveness, identifying with a product
- Wow in UX design

Why include emotional impact? Hassenzahl, Beu, & Burmester (2001, p. 71) and Shih and Liu (2007) put it this way: Users are no longer satisfied with just the efficiency and effectiveness of usability; they are also looking for emotional satisfaction. Norman (2004) uses more practical terms by stating that “attractive things make people feel good.” Users now seek pleasure in product use and the aesthetics in product design (Hassenzahl, 2012; Norman, 2002; Zhang, 2009). The products we own and use, especially high-tech, esoteric products, can arouse strong feelings of fun, importance, and social status.

Using an analogy from the discipline of architecture, building a house from a purely utilitarian perspective will likely result in a large box with walls and windows. The idea would be to optimize the space for providing shelter for the

largest number of occupants possible. In parts of the world where people are impoverished, this often tends to be the primary goal for design; however, in other parts of the world where people are more privileged, design of houses can incorporate empty spaces and other features that evoke beauty and delight. These spaces and features go beyond just shelter and match the functionality and structural amenities of the home to the lifestyles and values of the inhabitants.

What about fun interaction at work? Emotional impact factors, such as fun, aesthetics, and joy of use, are obviously desirable in personal use of consumer products, but what about in task-oriented work situations? Here, the need for usability and usefulness aspects of a user experience are obvious, but a positive emotional impact can also have value.

There is evidence that fun, or at least user fulfillment, can help break up monotonous work and can increase interest and attention span, especially for repetitive and possibly boring work such as that performed in call centers. Fun, increased user performance, and satisfaction from usage can enhance the appeal of less inherently challenging work (e.g., clerical work or data entry) (Hassenzahl, Beu, & Burmester, 2001). It is easy to see how user gratification and even fun can lead to job satisfaction and enjoyment of some kinds of work. Some kinds of empowerment also give users a real feeling of reaching the potential we have always expected from computers. An example is alignment lines for making diagrams in tools such as Keynote and OmniGraffle. This kind of support for intentionality, which is more than the contract of supporting a task sequence, can be satisfying.

Fun and usability can also conflict in work situations. Too simple can mean loss of attention, and consistency can translate as boring. Less boring, however, means less predictable, and less predictable usually goes against traditional usability attributes, such as consistency and ease of learning (Carroll & Thomas, 1988). Fun requires a balance: not too simple or boring, but not too challenging or frustrating.

Some work roles and jobs are simply not amenable to fun as part of the work practice. Consider a job that is inherently challenging and that requires full attention to the task, such as air traffic control. It is essential for air traffic controllers to have no-nonsense software tools that are efficient and effective. Any distraction due to novelty or even slight barriers to performance due to clever and interesting design features will be hated and could even be dangerous. For this kind of work, users need less mental effort, more predictable interaction paths, and more consistent behavior.

Deeper emotions. Most of the emotional impact factors are about pleasure, but they can also involve qualities such as love, hate, fear, mourning, and

reminiscing over shared memories. Applications in which emotional impact is especially important include social interaction (Dubberly & Pangaro, 2009; Rhee & Lee, 2009; Winchester, 2009) and interaction for cultural problem-solving (Ann, 2009; Costabile, Ardito, & Lanzilotti, 2010; Jones, Winegarden, & Rogers, 2009; Radoll, 2009; Savio, 2010).

The most basic reason for considering joy of use is the humanistic view that enjoyment is fundamental to life.

—Hassenzahl, Beu, & Burmester, 2001

Joy, excitement, and fun. We have adapted an example from Buxton's (2007b) book on sketching user experiences that illustrates the difference when emotional impact is a factor. Fig. 1.4 shows a picture of a mountain bike (similar to that of Buxton's [2007b, pp. 98–99]).

This bike is standing there ready for you to hop on and ride away on great adventures, but this image doesn't show that adventure, and that adventure is the user experience.

Now, contrast that with the image in Fig. 1.5 (again, like Buxton's [2007b, pp. 100–101]), which shows much more than the whole user (rider) or the whole bike.

The image shows the excitement of this user experience. The dynamic spray of water conveys the fun and exhilaration (and maybe a little danger). The blood and adrenaline are pumping as you careen over the bumpy rocks, and the scenery rushes by madly in a blur of motion. That is what you are buying—the breathtaking thrill of this biking experience.



*Fig. 1.4
A beautiful mountain
bike just waiting for
you to ride it.*



Fig. 1.5
The true mountain bike
experience.

Coolness and wow in UX design. These days, consumers are used to, and even expect, products that are really cool (Holtzblatt, 2011). Coolness and wow in the design are becoming required elements of emotional impact in the user experience (Hudson & Viswanadha, 2009).

Example: A Convincing Anecdote About the Importance of Emotional Impact

Pogue (2011) makes a convincing case for the role of emotional impact in a user experience via the iPad as an example. In his *New York Times* story, (Pogue, 2011) explains why the iPad turned the personal devices industry upside down and created a whole new class of devices. When the iPad came out, critics dubbed it “underwhelming,” “a disappointment,” and “a failure.” Why would anyone want or need it?

Pogue (2011) admits that the critics were right from a utilitarian or rational standpoint: “The iPad was superfluous. It filled no obvious need. If you already had a touch screen phone and a laptop, why on earth would you need an iPad? It did seem like just a big iPod Touch.” Yet as Pogue (2011) claimed, the iPad at that time was the most successful personal electronic device ever, selling 15 million in the first few months. Why? It has little to do with rational, functional, and utility appeal and has everything to do with emotional allure. It is about the personal experience of holding this artful object of cutting-edge technology in your hand and manipulating finely crafted objects on the screen. Once you have one, you find ways to make it useful.

Role of branding, marketing, and corporate culture. The emotional impact component of UX can extend to the milieu of what the manufacturer stands

for, how the product is marketed, and so on. What image does the brand of a product stand for? Is it a brand that uses environmentally sustainable manufacturing practices? What does the fact that someone is using a product of a particular brand say about them?

Consider the case of Apple in the late 2000s and early 2010s. The culture of designing for the user experience was so deeply engrained in their corporate ethos that everything they produced had a stamp of tasteful elegance and spectacular design. This kind of fanatic emphasis on quality user experience at Apple extended beyond the products they produced and seeped into other areas of their company. When they made an employment offer to a new employee, for example, the package came in a meticulously designed envelope that set the stage for what the company stood for (Slivka, 2009).

That aura pervaded Apple retail stores. A *New York Times* article (Hafner, 2007) extolled the enchanting aura of Apple stores: “Not only has the company made many of its stores feel like gathering places, but the bright lights and equally bright acoustics create a buzz that makes customers feel more like they are at an event than a retail store.” The goal of one new store in Manhattan was to make it “the most personal store ever created.” This carefully designed user experience has been very successful in generating sales, return visits, and even tourist pilgrimages.

Example: Branding and Passion for the Pontiac Car

There is an interesting story from General Motors about valuing (or not) branding and product passion. In October 2010, the board of directors quietly discontinued the Pontiac car from the GM line of brands. Of course, the direct cause was the transition through bankruptcy, but the beginning of the end for Pontiac started 26 years earlier.

Before that, Pontiac had its own separate facilities for design and production. Owners (and wannabe owners) were passionate about Pontiac cars, and Pontiac employees were devoted to the brand. The brand had its own identity, personality, and cachet, not to mention the notoriety from custom muscle cars such as the GTO and the Firebird Trans Am, as seen in the movie, *Smokey and the Bandit*.

In 1984, however, in its great corporate wisdom, GM lumped the Pontiac works in with its other GM facilities. The economically based decision to merge facilities meant no separate ideas for Pontiac design and no special attention to production. After that, there was no real basis for the devotion, and the passion was lost, leading to the decline and eventual demise of the brand.

1.4.4 Meaningfulness

Usability, and often emotional impact, is usually about a single usage occurrence; however, meaningfulness is, as it shows in [Fig. 1.3](#), about how a product or artifact becomes part of the life of a user. Meaningfulness comes out of a personal relationship of the product with its human user that endures over time. It is epitomized by the feelings of companionship many have for their smartphones, to the point that some users become physically uncomfortable if they become separated from their phones. Meaningfulness is exemplified by the feeling of comfort and safety felt by a hiker in response to a handheld global positioning system device.

1.5 EVOLVING IMPORTANCE OF UX

1.5.1 The Desire for Usability

Initial tolerance of a poor UX design. In the distant past, computer usage was esoteric, conducted mostly by a core of technically oriented users who were not only willing to accept the challenge of overcoming poor usability but also sometimes welcomed it as a barrier to protect the craft from uninitiated “outsiders.” Poor usability (Hartson, 1998) was good for the mystique, not to mention job security. For other users, being new to this incomprehensible technology meant acceptance of the learning and frustration ahead.

Dancing bear software. Sometimes, even more recently, we have what Cooper (2004, p. 26) calls “dancing bear” software, in which an amazing idea triumphs over poor design. Such products are about having features that are so good users cannot do without them, even if the software has a terrible interaction design. Having a bear that can dance leads one to overlook the fact that it cannot dance very well. Success despite poor interaction design can be used as a justification for resisting change and keeping the bad design ideas: “We have been doing it that way, our product is selling phenomenally, and our users love it.” Think of how much better it could be with a good design.

1.5.2 The Rise of Usability Engineering

Roughly in the 1990s, the focus moved to usability engineering, which focused heavily on elaborate usability evaluation methods. It has been almost three decades since the first usability engineering process books (e.g., Hix & Hartson, 1993; Mayhew, 1999; Nielsen, 1993) emerged from this period. This era encompassed early HCI concepts, but still had a bit of a software flavor (e.g., in the software tools for UI implementation) as well as standards for look and feel. This era of the discipline did much to improve usability of software everywhere and to raise awareness of the need to do so.

1.5.3 The Rise of UX

As we discussed earlier, usability remains an essential concept in HCI, but with the rebirth of Apple and the following democratization of good design, the focus shifted far beyond usability, addressing the totality of what a user feels when interacting with a product or system, including usability, usefulness, emotional impact, and meaningfulness. The skyrocketing importance of UX in not only the computer industry but in all walks of industrial life uncorked a genie in the bottle and made design of overarching importance.

Even staid old companies outside the technology industry began to recognize and embrace the need for UX as an indispensable part of their development practices. In an *Atlantic* magazine article about how Ford Motor Company hired Jim Hackett, a furniture designer, in 2019, as CEO, Useem (2019) explains, “We don’t live in the age of the automobile, or even the age of the computer. We live in the age of user experience.” Useem (2019) went on to say, “Into this arena has stepped a new class of professional: the UX designer, whose job is to see a product not from an engineer’s, marketer’s, or legal department’s perspective but from the viewpoint of the user alone.”

Even though it was not called UX at the time, this movement started in the early 1980s. Then, the personal computer put computing on a business user’s desk, and the consumer movement put computing in people’s houses. Customer service was the first to discover that expanding the marketplace to “mere mortals” without an adequate understanding of how the product was being used had a major impact on support costs.

Computers, however, and especially personal computers, were becoming more affordable to individuals, and it was more important for the average person to attain a good overall user experience during its use. Sometime about 2000, voices like that of Don Norman moved the focus from usability to a broader idea of the UX, embracing more of the total usage phenomenon,² and that’s where we are with this book!

1.5.4 Importance of Good UX Design and Costs of Bad UX Design

UX design has taken center stage in importance in product and system building. As a senior vice president of IBM said, “There’s no longer any real distinction between business strategy and the design of the user experience” (Kolko, 2015a, p. 70). Knemeyer (2015, p. 66) agrees, by saying, “User experience (UX) has become a mission-critical consideration for companies in every industry, and of every shape and size.”

²<https://www.nngroup.com/articles/definition-user-experience/>

One way to highlight the importance of good UX design is to cite the high cost of bad UX design. Bad UI/UX design can cost enormous amounts of money and, more importantly, lives. Distractions due to bad UX designs for operating cars can lead to traffic accidents, injuries, and death. The same caution applies to UX design for operating aircraft and ships at sea. For example, the fatal crash of EgyptAir Flight 990 in 1999 (see [Section 29.6.3.3](#)) was determined to be caused by poor usability in the design of cockpit controls. The collision of the USS *McCain* (Gallagher, 2017) is said to be the result of bad UX design of the navigation console.

In the medical domain, the need for good UX design is perhaps even more compelling with respect to the effects of safety in everyday operation. As Nielsen (2005) reports, “A field study identified 22 ways that automated hospital systems can result in the wrong medication being dispensed to patients. Most of these flaws are classic usability problems that have been understood for decades.”

1.6 SERVICE EXPERIENCE

Service experience and service design are specialized applications of UX (Forlizzi, 2010). As Forlizzi (2010, p. 60) says, service design is UX design for a user or customer experience within a “transactional journey.” It is about customer touch points in their user experience journey.

Documenting a service customer journey usually involves a storytelling narrative of a UX experience distributed over time and often distributed over different locations.

The customer journey is an abstraction of the main path. Users also experience deviations, edge cases, breakdowns, pinch points, and problems that crop up. Much of the UX research data gathering (see [Chapter 6](#)), analysis (see [Chapter 7](#)), and modeling (see [Chapters 8 and 9](#)) captures the mechanical steps of understanding this journey, including the emotional impact felt along the way.

Example: Service Experience in a Surgical Journey

Consider this hypothetical surgical service journey of a patient, describing how a health organization organized its services. Several pre-surgery hospital visits are choreographed by the hospital to fit their workflow, but they do not always seem to make sense for the patient. It begins with the family physician, who explores the patient’s symptoms and makes a tentative diagnosis. The doctor decides that further tests are needed to refine and confirm the diagnosis. So, off the patient goes to a local clinic, where they perform imaging (e.g., x-rays, MRI), confirming that surgery is required.

Next, the patient goes to a hospital in the nearest large city where the surgery is to occur. Getting there requires driving directions, which the clinic sends in an email. Upon arrival, the patient faces an overload of signage, including walking and entrance directions. The next job for the patient is to find the reception area to get directions to the appropriate department. Next is processing by a series of people who specialize in various pre-admission and preparation activities, and then self-entertainment while the patient waits to see a physician's assistant for the surgeon.

After all that, it is time for surgery—no instruction necessary. Post-surgical care, however, requires a sequence of prescriptions and trips to the pharmacy to fill them, and some phone calls to surgical staff at the hospital with questions about recovery. Several follow-up appointments are required with the surgeon and then with the family physician, all requiring the patient to maintain a calendar of events. This is quite a customer journey and is fraught with potential UX problems at every turn.

1.7 WHY SHOULD WE CARE? THE BUSINESS CASE FOR UX

Ingenious by design; hassle-free connectivity

—On a Toshiba satellite receiver box

If you do not have a convincing business case for UX, then you really have to conclude that, although it might be interesting, at the end of the day, UX is simply an academic exercise.

1.7.1 Is the Fuss Over UX Real?

In the past, this might have been a question usability practitioners had to face: Is the fuss over UX real? UX is now recognized as a key job role in product development and a key part of the development process. We no longer have to defend UX and usability because UX has penetrated most corporate and organizational development groups and projects. The success of “design thinking” (Brown, 2008) in business (the mindset of getting organizations to think like designers, applying design principles and practice to businesses and business processes) is one reason, and the rise of Apple is another.

1.7.2 Cost Justification

In its early days, usability often failed to garner support, especially from people in management. Usability engineers felt obligated to justify their existence and prove the value of their work. They did it by cost-justifying usability (Bias & Mayhew, 2005; Mantei & Teorey, 1988), which involved examples of how much money a design improvement could save, given the time it took to perform a transaction involving the old design and the new design and the frequency by which the transaction occurred. That kind of cost-benefit analysis was good, as far as it went, but realistically, other factors were more compelling; customers and users were becoming more computer sophisticated and less tolerant of poor interaction designs. Marketing people were becoming more aware of the demand for good design and getting into the act by demanding good design from their own organizations.

So, these days, the idea of cost-justifying UX design work is out of date. No one asks for cost justification on the software engineering side, and the same is becoming true on the UX side. Managers are realizing that it pays off in many ways to focus on getting a product or system right the first time. The result will be less total time, less money, and less negative impact on customer goodwill.

1.8 HISTORY RELATED TO HCI AND UX

We wrap up this chapter with a terse capsule of the history and roots of HCI, usability, and UX. The coverage is by no means a survey of the vast contributions in this large field.

It is a matter of debate exactly when the notion of computer usability was born. In many ways, it was a by-product of the introduction of personal computers. We know that it was a topic of interest to some by the late 1970s, and by the early 1980s, conferences about the topic were being established. Although work was being done on “human factors in computers” in the 1970s and earlier, HCI was emerging at various universities in the late 1970s and 1980s and had been going on at IBM (Branscomb, 1981), the National Bureau of Standards (now the National Institute of Standards and Technology), and other scattered locations before that.

Many believe that HCI did not coalesce into a fledgling discipline until the first Conference on Human Factors in Computer Systems (CHI) in Boston in 1983, but it probably began at least a couple of years before with the “unofficial first CHI conferences” (Borman & Janda, 1986) at (1) the May 1981 Association for Computing Machinery (ACM)/Special Interest Group for Social and Behavioral Science Computing (SIGSOC) conference, called

the Conference on Easier and More Productive Use of Computer Systems, in Ann Arbor, Michigan, and (2) the March 1982 Conference on Human Factors in Computer Systems in Gaithersburg, Maryland. A more detailed account of the history of the ACM SIGCHI (Special Interest Group for Human-Computer Interaction) conferences can be found in Kumar et al. (2022).

HCI, in general, and usability and UX, in particular, owe much of their origin and development to influences from many other related fields.

1.8.1 Early Influence of Industrial, Human-Factors Engineering, and Ergonomics

By the time the United States was engaged in World War I, training was used to fit the human to work with the machine. Engineers built aircraft with all their controls and displays and then brought in potential pilots and trained them to fit the designs. With the era of World War II, a new discipline, called human factors and ergonomics, was emerging. Early heroes in this new field included Col. Paul Fitts, Capt. Richard Jones, and Lt. Alphonse Chapanis. US Air Force officials were concerned about airplane crashes suffered by experienced World War II pilots. Very fit, highly trained, and highly motivated people were making significant task performance errors in the field. No matter how extensive the training or flying experience, pilots were making dangerous mistakes while operating controls in the cockpit. Well-trained pilots were failing to detect enemy on radars and were making “pilot errors” in airplane cockpits.

Fitts and Jones (1947) concluded that the cause could not be flaws in training and were among the very earliest that recognized the causal connection between design flaws, rather than human errors, and mishaps in user performance. So, they studied “critical incidents” that may have led to airplane crashes. They interviewed pilots about critical incidents they experienced, a practice that lives on today in HCI and UX.

To continue this theme in the mid-1950s, Henry Dreyfuss, an American industrial designer, wrote in his seminal book, *Designing for People*, that difficulties in interaction with a product are not necessarily the fault of the human user, but often it is because the designer has failed to appropriately support the needs of those users (Dreyfuss, 1955).

1.8.2 Human Factors, Psychology, and Cognitive Science Meet HCI

It is not surprising that human factors, an engineering discipline bringing science and technology together with human behavior, and biological characteristics for design fit well with the emerging field of HCI and usability, laying the basis for HCI techniques later on.



Fig. 1.6
The first computer mouse
(Apic/Hulton Archive via
Getty Images).

Much of the foundation for HCI has also been closely related to theory in psychology, derived from adaptations of psychological theory and models of human information processing (Barnard, 1993; Hammond, Gardiner, & Christie, 1987). Concepts such as user modeling and user performance metrics were adopted into HCI from cognitive and behavioral psychology and psychometrics.

1.8.3 Computer Science: Hardware and Software Foundations of HCI

In the 1960s, more people with less technical training were getting into contact with computers. Until this time, all user interaction was keyboard based—until the very first experimental pointing device, the first mouse (Fig. 1.6), was invented by Douglas Engelbart.³

A new focus in computer science emerged in creating devices, interaction styles, and supporting software for interoperability and human ergonomics. Soon (ca. 1960), we were in an era of research and development on keyboards, cathode-ray tube terminals, text editors, and training manuals. At this point, the main participants in HCI from computer science were software people, and much of the work was about input/output devices and UI programming.

1.8.4 Interactive Graphics, Devices, and Interaction Techniques

A new thread began appearing in the literature and practice on the computer science side of the HCI equation. This work on interactive graphics,

³https://en.wikipedia.org/wiki/Computer_mouse

interaction styles, software tools, dialogue management systems, programming language translation, and interface widgets was essential in opening the way to practical programming techniques for bringing interaction designs to life on computers (Myers & Olsen, 1994).

The origin of computer graphics is frequently attributed to pioneers such as Ivan Sutherland (1963a, 1963b) and solidified by masters such as Foley and colleagues (Foley & Van Dam, 1982; Foley et al., 1990; Foley & Wallace, 1974) and Newman (1968). For an insightful account of the relationship of computer graphics to HCI, see Grudin (2006).

The 1980s and 1990s saw a burgeoning of hardware and software development to support the now familiar point-and-click style of interaction, including the Xerox Star (Smith et al., 1989) and the Lisa and Macintosh by Apple. Personal computing brought with it a kind of democratization of computing—computing for everyone, not just the elite geeks.

In the context of an interaction technique (Foley et al., 1990), an interaction object and its supporting software were often referred to as a widget. Programmers developed libraries of widget software to support programming of GUIs. Early graphics packages took interaction beyond text to direct manipulation of graphical objects, eventually leading to new concepts in displays and cursor tracking. No longer tied to just a keyboard or even just a keyboard and mouse, many unusual (then, and some still now) interaction techniques arose (Buxton, 1986; Hinckley et al., 1994; Jacob, 1993). Myers led the field in UI software tools of all kinds (Myers, 1989, 1992, 1993, 1995; Myers, Hudson, & Pausch, 2000), and Olsen (1983) is known for his work in UI software.

So many people contributed to the work on User Interface Management Systems (UIMS) that it is impossible to recognize them all. Buxton and colleagues (1983) were among the earliest thinkers in this area. Others we remember are Brad Myers, Dan Olsen, Mark Green, Jim Foley's group at GWU, and our researchers at Virginia Tech. Much of this kind of work was reported in the ACM Symposium on User Interface Software and Technology, a conference specifically for UI software.

The commercial world followed suit, and we worked through quite a number of proposed standard interaction styles, such as OSF Motif (The Open Group). Developers had to choose from those available mainly because the interaction styles were not interoperable. Each approach was tied closely to its own software tools for generating the programming code for interaction designs using the devices and interaction styles of these approaches. Standardization led to today's GUI platforms and corresponding styles.

This growth of graphics and devices made possible one of the major breakthroughs in interaction styles—direct manipulation (Hutchins, Hollan, & Norman, 1986; Shneiderman, 1983), changing the basic paradigm of interaction with computers. Direct manipulation allows opportunistic and incremental task planning. Users can try something and see what happens, exploring many avenues for interactive problem-solving.

1.9 SEGUE

This chapter is a brief introduction to UX, what it entails, why it is important, and how we arrived at this point in time as a discipline. In the next chapter, we discuss The Wheel, our framework for organizing UX processes, lifecycles, methods, and techniques.